THARUN KUMAR KSHEERASAGAR

Senior Architect

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SUMMARY

System Architecture and Performance Evaluation Engineer with over a decade driving pre-silicon validation and compute acceleration at NVIDIA, AMD, and Samsung. Expertise in hardware-software co-design, hybrid simulation, and optimizing complex compute subsystems to eliminate system bottlenecks. Recent work spans advanced AI infrastructure, from integrating NVIDIA iGPUs into MediaTek SoCs to performance modeling for next-gen accelerated inference systems.

WORK EXPERIENCE

Stealth AI Hardware Startup (Operations Ceased) — Hyderabad, India
Senior Architect — *Apr 2026 – Sept 2026*

- Led a two-engineer team while developing a configurable SystemC performance model delivering exact cycle-by-cycle execution for a Groq-inspired AI inference accelerator card tailored for Transformer and LLM workloads.
- Provided a golden reference model for the compiler team, establishing predictable, low-latency execution frameworks for real-time Transformer workloads.
- Prototyped an intuitive web-based data-flow visualization tool using agentic AI to track real-time packet movement across hardware blocks, accelerating compiler debug turnaround times.

NVIDIA — Hyderabad, India
Senior Architect — *Jan 2025 – Apr 2026*

- Led system-level validation and cross-functional debug for NVIDIA's next-generation N2X AI platform.
- Spearheaded the complex integration of an NVIDIA iGPU into a MediaTek SoC, driving virtual platform bring-up to achieve successful system Linux and GPU firmware boot in an accelerated 4-month timeframe.

AMD - Xilinx — Hyderabad, India
Member of Technical Staff Software Engineer — *June 2019 – Jan 2025*

- Designed and delivered a custom multi-threaded simulation kernel within a strict six-month timeframe, maintaining SystemC semantics while accelerating simulation speeds by up to 10x.
- Drove the integration of AMD's AI Engine (AIE1/AIE2) Neural Processing Unit (NPU) models into the new infrastructure, exceeding the initial proof-of-concept scope by transpiling and fine-tuning a cycle-accurate model into a functional, untimed simulation model.
- Enhanced and validated cycle-accurate performance models for AMD AI Engine (AIE1/AIE2) stream switches and Memory Tiles (Mem Tiles) to enable precise component-level performance evaluation.
- Developed a comprehensive software debug framework tailored for AI workloads, providing robust tooling to troubleshoot and optimize Convolutional Neural Network (CNN) architectures such as ResNet and YOLO.
- Developed a custom memory model in Xilinx Vitis for advanced performance analysis of Alveo data center accelerator cards, supporting high-performance compute workloads across Machine Learning (ML) inference and FinTech algorithmic trading.
- Accelerated Xilinx Vitis hardware emulation speed by 6x (doubling the MVP target of 3x), significantly reducing simulation and debug turnaround times for complex architectures.
- Engineered a scalable, latency-accurate I/O emulation feature for the Xilinx Accelerated Algorithmic Trading (AAT) framework, enabling cycle-accurate simulations using PCAP inputs for precise packet-level visibility.

SAMSUNG — Bengaluru, India
Associate Staff Engineer — *June 2016 – June 2019*

- Architected a Cycle-Accurate Hybrid Simulation Framework that synchronized SystemC and FPGA environments, enabling seamless emulation of complex, distributed designs.
- Developed critical cycle-accurate models and transactors, including an AXI3 Interconnect SystemC model and AXI-to-TLM transactors, to drive advanced performance measurement for memory subsystems.
- Engineered a custom Co-Simulation framework leveraging Interconnect WorkBench (IWB/Incisive), SystemC, and NCSim to streamline cross-platform verification and design testing.

PATENTS

- **Model Level Debugging of Machine Learning Designs on Neural Processing Units** (Published, US-20260186951-A1)
- **Multi-Threaded Cycle Accurate Architecture Simulation** (Published, US20240184616-A1)
- **Data traffic injection for simulation of circuit designs** (Issued, US-11630935-B1)
- **Circuit design simulation and clock event reduction** (Issued, US-12086521-B2)
- **Adaptive Simulation Framework for Switching Between Functional and Cycle-accurate Execution in Real-Time** (Pending)

PUBLICATIONS & PRESENTATIONS

- **Performance analysis of DS-CDMA using different chaotic sequences** – IEEE, 2016 Int. Conf. on Wireless Communications, Signal Processing and Networking (WiSPNET), Mar 2016, pp. 2421–2425
- **Multi-Threaded Simulation Modeling Kernel** – Poster Presentation, AMD SW Innovation Expo (internal), 2023

CERTIFICATIONS

- **GPU Architectures and Programming** – NPTEL (Topper) *May 2022*
- **Multi-Core Computer Architecture – Storage and Interconnects** – NPTEL (Elite Gold) *Oct 2022*
- **Operating System Fundamentals** – NPTEL (Topper) *Nov 2021*
- **Modern C++ Concurrency in Depth** – Udemy *Oct 2020*

SKILLS

- **Languages & Libraries:** C++, SystemC, Python, Verilog
- **Simulation & Modeling:** Cycle-accurate simulation, performance modeling, architectural exploration, microarchitecture
- **Tools & Frameworks:** Synopsys Virtualizer Studio, AMD Vivado / Vitis, AMD xsim, Cadence IWB / Incisive, NCSim, FPGA emulation/prototyping
- **Domains:** AI/accelerator cards, NPU and AI accelerator microarchitecture, Transformers, LLMs

AWARDS

- **AMD:** Executive Spotlight Award, ACE Next 5% Award
- **Xilinx:** Quarterly Award, multiple Boost awards, multiple SPARK awards
- **Samsung:** multiple Employee of the Month awards, multiple Spot awards, Professional Competence Certification (internal coding competency test)

EDUCATION

M.Tech in Software Systems BITS Pilani (WILP), (2018 – 2020)
CGPA: 8.19

B.Tech in Electronics & Communication Engineering NIT Warangal, (2012 – 2016)
CGPA: 8.94 | Consistent Institute Merit Scholar